Mahankali Praharsha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.23
NARAYANA JUNIOR COLLEGE	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2023	98.40%
JAYA HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION , TELANGANA)	2021	10.00%

PROJECTS

Reinforced Concrete Structural Design & Detailing

- **Designed** reinforced concrete **slab and beams** by evaluating load effects, bending moments, and reinforcement requirements, developing Practical structural detailing in accordance with relevant design provisions.
- **Analyzed** structural response under **dead and live load combinations** using STAAD.Pro, effectively identifying governing design actions to support subsequent member design and ensure efficient structural assessment.
- **Developed** reinforcement detailing for structural members, incorporating **flexural and shear design, reinforcement curtailment, and development-length checks** to ensure adequate detailing and code compliance.

Comparative Assessment of IS 456:2000 and Draft Code Provisions

- **Analyzed** changes in the **concrete and steel stress-strain curves, stress-block parameters, limiting neutral-axis depth, and minimum tension reinforcement** across **IS 456:2000** and Draft Code provisions.
- **Evaluated** singly and doubly reinforced **RCC sections** through the flexural design, determining limiting moment capacity and reinforcement requirements using the **force-moment equilibrium** under both code provisions.
- **Developed** an iterative approach using the **strain compatibility and force-moment equilibrium** to assess the concrete moment capacity and compare the axial-moment response using column interaction diagrams.

Gateway of India - 3D Modeling & Printing

- **Designed** a detailed 3D CAD model of the Gateway of India using the **Solid Edge**, applying engineering drawing principles and coordinating model development for **precise prototyping and effective presentation**.
- **Collaborated** with in a team to execute design and presentation, **enhancing teamwork and technical skills.**
- **Developed** components and assemblies, translating engineering **drawings into** accurate **3D CAD** representation.

LST Data Gap Analysis and Spatial-Temporal Interpolation

- **Analyzed** a **25-year LST dataset (2001-2025)** by assessing missing-data patterns and structuring spatio-temporal raster data across the spatial and temporal dimensions for interpolation and reconstruction analysis.
- **Implemented temporal and spatial interpolation techniques**,including linear interpolation and IDW-based gap filling,simulating temporal,spatial,and,random gaps and evaluating **reconstruction performance through metrics**.
- **Evaluated** interpolation performance across different gap scenarios and parameter settings, identifying suitable approaches for **reconstructing missing LST observations and improving spatial-temporal data continuity**.

Development of an Integrated Air Quality and Climate Risk Prediction Tool Using ML (Ongoing)

- **Compiled** multi-year **air-quality and meteorological data (2021-2026)** from monitoring stations, integrating pollutant and the climate parameters to establish a structured dataset for the environmental risk analysis.
- **Processed and analyzed** key **air-quality and climate variables**, including computation of Air Quality Index **(AQI)** and Wet Bulb Globe Temperature **(WBGT)** to characterize combined air-pollution and thermal-stress conditions.
- **Developed** an integrated **Environmental Health Risk Index (EHRI)** and evaluated **machine-learning models** using k-fold cross-validation to classify environmental health-risk categories from climate and air-quality indicators.

SKILLS

Technical Skills: Python-familiar with NumPy, Pandas, Matplotlib, Seaborn for data analysis and visualization

Tools & Soft Skills: Microsoft Office, QGIS, Solid Edge, Critical and Logical Thinking, Problem-Solving, Teamwork

RELEVANT COURSES

Mathematics courses: Probability, Statistics, Complex Variables, Elementary Linear Algebra, DE, Calculus

Core Courses: Construction Materials and Management, Mechanics of Solids and Structures, Structural Analysis, Environmental Systems Engineering, Geotechnical Engineering, Surveying, Fluid Mechanics, Hydraulic Engineering, Foundation Engineering, Reinforced Concrete Design, Engineering Hydrology, Highway Design and Materials, Water Resources Engineering, Traffic Engineering and Planning, Railway and Airport Engineering.

EXTRACURRICULAR

- **Achieved multiple medals in Chess** at school, inter-district, state-level, and IIT Hyderabad competitions, demonstrating strategic thinking and focus and Competitive spirit.
- **Represented** NSO Football at IIT Hyderabad, actively participating in team activities and **demonstrating teamwork.**
- **Interested in creative and personal activities** such as music, dance, reading newspapers, and watching films, developing creativity, cultural awareness, and a balanced range of personal interests.

- **Contributed** to the **Learn In Bharat (LIB) Program 2026** organized by **Next Bharat Ventures**, supporting workshop execution and actively participating in program activities.